SECURE DEVICE REGISTRATION & INTELLIGENT ACCESS CONTROL SYSTEM

SOA Project Review-3 – Final Review

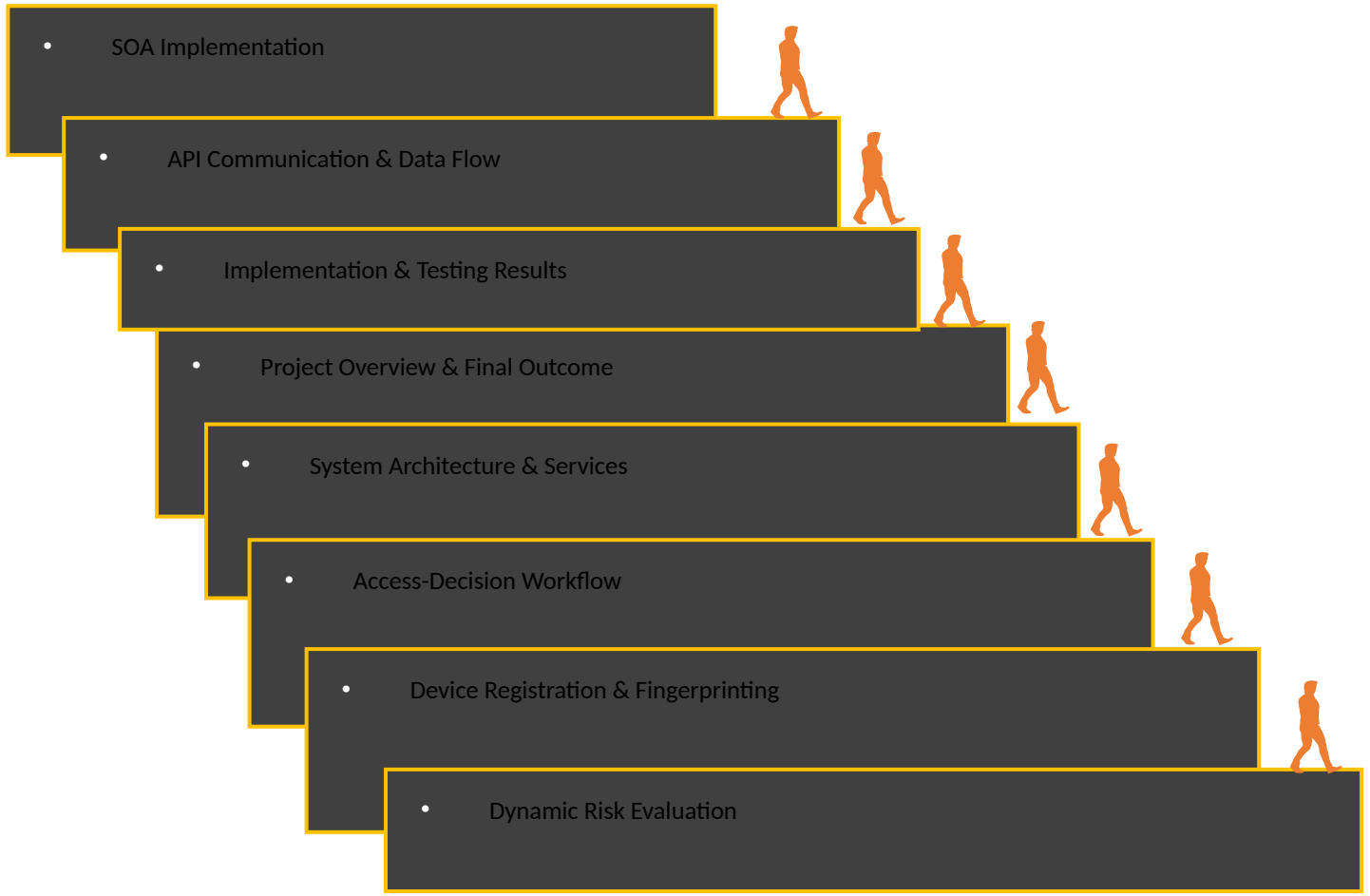
Final Project Review

Under the Guidance of

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Final Review: Project Completion



Project Overview & Final Solution

Our project adds a device-aware security layer to access control. Instead of relying on a password alone, it considers whether the device is registered and active, whether the user is authenticated, and how risky the requested action appears.

The completed workflow brings together:

- Device registration and fingerprint generation
- User authentication and device-status verification
- Context-based risk scoring and suspicious-activity checks
- Policy-based access decisions and audit logging

The result is a clearer, traceable way to allow or block requests and help administrators investigate unusual activity.

Completed Objectives

1) • Built a device registration and management workflow.
• Added device fingerprint generation and verification.
• Connected device records with user identity for access checks.
• Implemented a risk-scoring approach with scores from 0 to 100.
• Added checks for suspicious access patterns and repeated failures.
• Implemented policy-based ALLOW / BLOCK decisions.
• Recorded access outcomes and security events with timestamps and device details.
• Created a central dashboard for reviewing devices, risk, alerts, and recent events.
• Organised the system around API-based services to support the SOA approach.

Research Foundation & Design Decisions

Authors / Area	Title / Approach	Methodology	Key Inference	Limitation / Future Work
Device Fingerprinting	Device identity & risk-based access	Device attributes and behaviour can help distinguish devices and estimate trust.	A device fingerprint adds another signal to user authentication.	Some attributes change over time, so fingerprints need careful design.
Rose et al. / NIST	Zero Trust Architecture	Zero Trust verifies each request instead of assuming a user or device is safe.	Access should be checked continuously, not granted on identity alone.	Applying these ideas at scale requires careful service and policy design.
Dynamic Risk / IoT Fingerprinting	Context-aware device access	Contextual risk and fingerprinting can be combined for device-aware decisions.	Risk context helps the system respond differently to normal and unusual requests.	Risk weights, data quality, and scalability need ongoing evaluation.

Final Access-Control Workflow

1. Device registration: store the device record and generated fingerprint.

2. Authentication: validate the user's credentials before protected access.

3. Device verification: check whether the fingerprint is recognised and the device is active.

4. Risk evaluation: combine signals such as device trust, failed attempts, behaviour, and resource sensitivity.

5. Access decision: apply role and resource policies alongside the risk result to ALLOW or BLOCK the request.

6. Audit and monitoring: record the decision, timestamp, device details, and reason so the administrator can review it.

This integrated flow brings user identity, device trust, contextual risk, and monitoring together.

Completed Features & Key Contributions

What we implemented

- Device registration, fingerprint hashing, and registry storage.
- Dynamic risk scoring with four tiers: Low (0–25), Moderate (26–50), High (51–75), and Critical (76–100).
- Access decisions based on device status and risk, with enforcement and audit records.

How it helps

- Gives administrators a consolidated view of devices, risks, alerts, and access events.
- Makes denials easier to understand by keeping a reason and event history.
- Separates major responsibilities into API-based services for maintainability.

Key contribution

The project combines device trust, user identity, contextual risk, and resource sensitivity in one access-decision workflow. Fingerprinting is treated as a useful recognition signal, not an infallible hardware identity.

Implementation, Testing & Final Outcome

Completed implementation

- Device registration pipeline: fingerprint capture → hash generation → registry storage.
- Dynamic risk scoring engine with a 0–100 score and defined risk tiers.
- Access enforcement for ALLOW / BLOCK outcomes.
- API-based SOA components and a security dashboard for reviewing activity.
- PostgreSQL audit logging for traceable security events.

Testing carried out

- Checked fingerprint extraction and hash consistency across 10 device profiles.
- Validated the end-to-end device registration flow.
- Confirmed risk scores are generated within the 0–100 range.
- Verified that ALLOW / BLOCK outcomes can be recorded with timestamps and device details.

Final outcome & next steps

The project demonstrates a device-aware access-control workflow. Further work can improve ML anomaly detection with larger datasets, handle IP changes more smoothly, expand cross-platform support, and conduct a formal security audit.